

CLOUD COMPUTING COURSE

Lecture25- Cloud Storage

ABSTRACT
Lecture25 Introduces cloud storage.

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Cloud Storage

Cloud storage Definition

Cloud storage is a cloud computing model in which data is stored on remote servers accessed from the internet, or "cloud." It is maintained, operated and managed by cloud storage service provider on storage servers that are built on virtualization techniques.

Cloud storage works through data center virtualization, providing end users and applications with a virtual storage architecture that is scalable according to application requirements.

Creating Cloud Storage System

The cloud storage system stores multiple copies of data on multiple servers, at multiple locations. If one system fails, then it is required only to change the pointer to the location, where the object is stored.

In cloud computing, the virtualization program formats the LVs.

Vendors use different virtual file systems. For example**, VMware** uses Virtual Machine File System **(VMFS),** and Huawei uses Virtual Image Manage System **(VIMS).** Both of them are high-performance cluster file systems that deliver a capacity exceeding the limit of a single system and allow multiple compute nodes to access an integrated clustered storage pool.

The minimum storage unit used by virtualization programs **is logical unit number (LUN).**

LUNs correspond to volumes. Each LUN represents an individual storage volume (it acts as an identifier for a disk volume in the storage).

Storage Resource Pool: is a pool that pools the physical storage from different network storage devices and makes it appear to be a single storage unit that is handled from a single console.

After virtualization is used, **LUNs can be divided** as:

1. **Thick LUNs**
2. **Thin LUNs.**

	Thick LUNs	**Thin LUNs**
Capacity	- Once being created, get full storage capacity from the storage pool. - Less storage capacity utilization	- Get storage capacity on demand. - Higher storage capacity utilization
Disk space reclamation	**does not apply** to a thick LUN (if data in a thick LUN is deleted, the allocated capacity is occupied by the thick LUN and cannot be used by other LUNs)	When data in a thin LUN is deleted, space in the thin LUN **can be released.**
Performance	A thick LUN delivers **higher performance for sequential** reads/writes as it gets full storage capacity from the beginning, but it has some storage capacity wasted.	A background formatting is required each time the thin LUN expands capacity, In addition, capacity allocations may cause discontinuous disk storage space, so it **takes more time** for sequential reads/writes to find storage locations. Which **cause less performance**.
Application scenarios	− High performance is required. − Storage space utilization is less concerned. − Costs are insensitive.	- Moderate performance is required. − Storage space utilization is more concerned. − Costs are sensitive. − Required storage capacity is hard to predict.

VM Disks

A VM consists of **configuration files and disk files**. Each VM disk corresponds to a disk file where **user data is stored**.

If **virtualized** storage is used, all disk files are stored in the **shared directory** of the file system.
If non-virtualized storage is used, each **disk file corresponds to a LUN**.

From the perspective of users and OSs, either files or LUNs are the same as common hard drives, which are displayed as hard drives among the hardware resources of the system.

When creating a VM, the administrator needs to create disks for the VM to store data. The disk information corresponds to several lines in the configuration file.

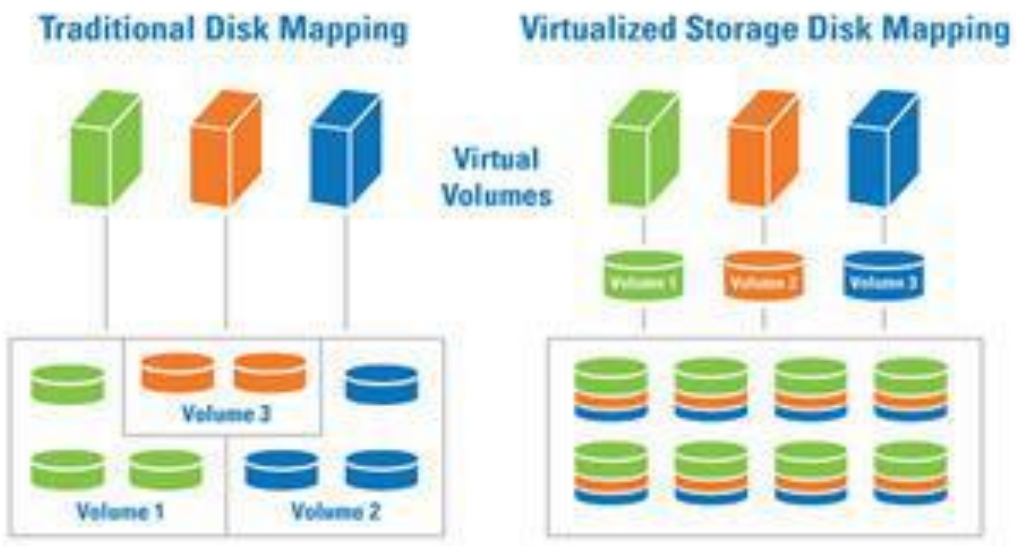
VM disk files have their own fixed formats and there is many common VM disk format as shown in the table:

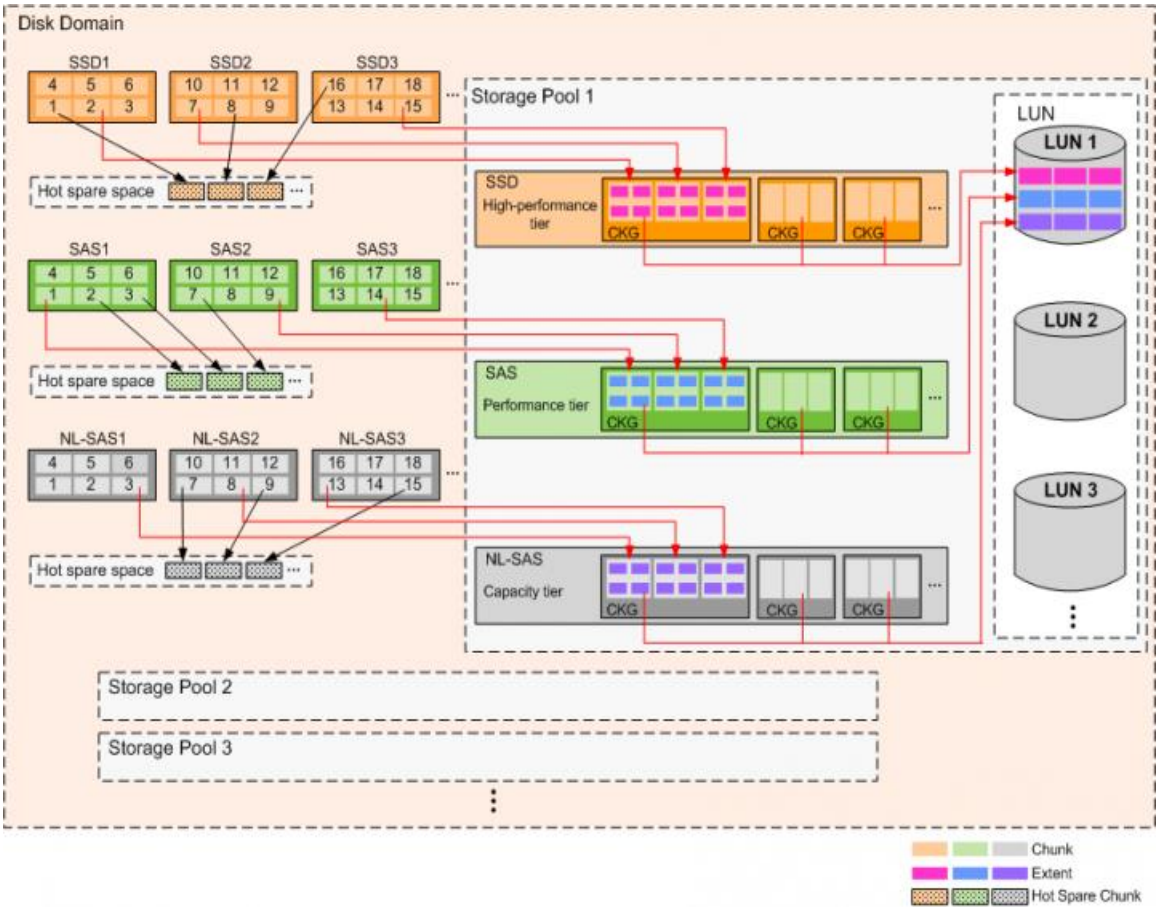
Common VM disk formats

VM Disk File Format	Supported Vendor, Product, or Platform
RAW	All vendors
VMDK	VMware
VHD	Microsoft Hyper-V and Huawei FusionCompute
QCOW QED	QEMU or KVM virtualization platforms
VDI	Oracle

Each vendor can use **its own tool to convert other VM disk formats** to formats that can be used by its own products.

For example, Huawei Rainbow can convert third-party or open-source VM disks to the VHD format.


Traditional Disk Mapping
Virtualized Storage Disk Mapping
Virtual Volumes
Volume 3
Volume 1
Volume 2


Disk Domain
SSD1
SSD2
SSD3
Hot spare space
SAS1
SAS2
SAS3
Hot spare space
NL-SAS1
NL-SAS2
NL-SAS3
Hot spare space
Storage Pool 1
SSD
High-performance tier
SAS
Performance tier
NL-SAS
Capacity tier
CKG
LUN
LUN 1
LUN 2
LUN 3
Storage Pool 2
Storage Pool 3
Chunk
Extent
Hot Spare Chunk